

EE2012 Cheat Sheet

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1 Prob Basics & Combinatorics

1.1 Set Operations & Axioms

De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$

Axioms: $P(A) \geq 0$; $P(S) = 1$;

$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$.

Properties:

• $P(A^c) = 1 - P(A)$; $P(\emptyset) = 0$; If $A \subset B \rightarrow P(A) \leq P(B)$

• $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

• $P(A \cup B \cup C) = \sum P(A) - \sum P(AB) + P(ABC)$

1.2 Counting Principles

• **Permutation (Order matters):** $P_k^n = \frac{n!}{(n-k)!}$ (no replace); n^k (replace)

• **Combination (Order doesn't matter):** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ (no replace)
Variance of Binomial = $np(1-p)$. S.D. (σ) = $\sqrt{\text{Variance}}$.

• **Stars & Bars (Replace, no order):** $\binom{n+k-1}{k}$ items into bins

1.3 Conditional Probability & Independence

Def: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. **Chain Rule:** $P(AB) = P(A|B)P(B)$.

Independence: $P(AB) = P(A)P(B)$ or $P(A|B) = P(A)$.

Note: Disjoint ($P(AB) = 0$) $\neq$ Independent (unless prob=0).

1.4 Total Probability & Bayes' Theorem

Let $B_1, \dots, B_n$ be a partition of S :

• **Total Prob:** $P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$

• **Bayes' Rule:** $P(B_k|A) = \frac{P(A|B_k)P(B_k)}{\sum P(A|B_i)P(B_i)} = \frac{P(A|B_k)P(B_k)}{P(A)}$,
where $P(A) = P(A|B_k)P(B_k) + P(A|B_i)P(B_i)$

1.5 Sequential Experiments

Independent: $P(A_1 \dots A_n) = P(A_1) \dots P(A_n)$

Markov Chain: $P(A_n|A_{n-1} \dots A_1) = P(A_n|A_{n-1})$

2 Discrete Random Variables

PMF: $p_X(x) = P(X = x)$. $\sum p_X(x) = 1$.

CDF: $F_X(x) = P(X \leq x) = \sum_{k \leq x} p_X(k)$ (Staircase).

Expectation: $E[X] = \sum_x xp_X(x)$. $E[g(X)] = \sum g(x)p_X(x)$.

Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$ $\text{Var}(X) = np(1-p)$.

Mean: $\text{Mean } E[X] = np$. $E[X + Y] = E[X] + E[Y]$ (Linearity).

Properties:

• $E[aX + b] = aE[X] + b$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$

• $E[X + Y] = E[X] + E[Y]$ (Always)

• $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ (If Independent)

2.1 Common Discrete Distributions

Dist	PMF $P(X = k)$	Mean	Var
Bernoulli(p)	$p^k(1-p)^{1-k}, k \in \{0, 1\}$	p	pq
Binomial(n, p) $k = 0, \dots, n$	$\binom{n}{k}p^k(1-p)^{n-k}$ <i>Sum of n i.i.d Bernoullis</i>	np	npq
Geometric(p) $k = 1, 2, \dots$	$(1-p)^{k-1}p$ <i>Trials until 1st success. Memoryless.</i>	$1/p$	$(1-p)/p^2$
Neg. Binom(r, p) $k = r, r+1, \dots$	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$ <i>Trials until r-th success.</i>	r/p	$r(1-p)/p^2$
Poisson(λ) $k = 0, 1, \dots$	$e^{-\lambda} \frac{\lambda^k}{k!}$ <i>Events in interval. Lim of Binomial.</i>	λ	λ
Uniform(a, b)	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$

3 Continuous Random Variables

PDF: $f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x)dx = 1$

PDF From CDF: $f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}$

Prob: $P(A) = \int_A f_X(x)dx$. $P(X = c) = 0$.

CDF: $F_X(x) = \int_{-\infty}^x f_X(t)dt \iff f_X(x) = \frac{d}{dx} F_X(x)$.

Expectation: $E[X] = \int_{-\infty}^{\infty} xf_X(x)dx$.

Symmetry: If symmetric about m , $E[X] = m$.

Non-negative RV: $E[X] = \int_0^{\infty} (1 - F_X(x))dx$.

3.1 Common Continuous Distributions

Dist	PDF $f_X(x)$	Mean	Var
Uniform(a, b)	$\frac{1}{b-a}, a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ) <i>CDF: $1 - e^{-\lambda x}$</i>	$\lambda e^{-\lambda x}, x \geq 0$ <i>Inter-arrival time of Poisson</i>	$1/\lambda$	$1/\lambda^2$
Gaussian(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Rayleigh(σ^2)	$\frac{r}{\sigma^2} e^{-r^2/2\sigma^2}, r \geq 0$	$\sigma\sqrt{\frac{\pi}{2}}$	$\frac{4-\pi}{2}\sigma^2$

3.2 Gaussian (Normal) Properties

Standard.: $X \sim N(\mu, \sigma^2) \rightarrow Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$. $\sigma_{new} = \frac{\sigma}{\sqrt{N}}$

Mean: np **Var:** $\sigma^2 = np(1-p)$

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

Q-Function: $Q(z) = P(Z > z) = \int_z^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$.

• $Q(0) = 0.5$, $Q(-z) = 1 - Q(z)$, $\Phi(z) = 1 - Q(z)$

• $P(|Z| > c) = 2Q(c)$

Sum: Independent $X_i \sim N(\mu_i, \sigma_i^2)$:

$$\sum a_i X_i \sim N\left(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2\right)$$

Standard Normal Distribution (Positive Z-Scores)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

4 Functions of Random Variables

Let $Y = g(X)$. **Method 1: CDF (General)** 1. Find support of Y .

2. $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$. 3. Integrate $f_X(x)$ over region

$\{x : g(x) \leq y\}$. 4. $f_Y(y) = \frac{d}{dy} F_Y(y)$. **Method 2: Transformation**

Formula (Monotonic)

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \text{ where } x = g^{-1}(y)$$

Linear: $Y = aX + b \Rightarrow f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$. **Sum**
(Convolution): $Z = X + Y$ (Indep)

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

5 Joint Distributions

Joint PDF/PMF:

• Discrete: $p_{XY}(x, y) = P(X = x, Y = y)$

• Continuous: $P((X, Y) \in D) = \iint_D f_{XY}(x, y) dx dy$

• Normalization: $\iint f_{XY} dx dy = 1$

Marginals: $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$ **Conditional:**

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} \quad \textbf{Independence: } f_{XY}(x, y) = f_X(x)f_Y(y)$$

Visual check: Region must be rectangular ($a < x < b, c < y < d$) AND density factorizable. If region is triangle ($0 < y < x$), NOT independent.

5.1 Covariance & Correlation

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Uncorrelated if $\text{Cov}(X, Y) = 0$ (Indep $\Rightarrow$ Uncorr)
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

$$\textbf{Correlation: } \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

5.2 Complex Gaussian (EE Special)

Let $X, Y \sim N(0, \sigma^2)$ be independent. **Complex RV:**

$H = X + jY \sim \mathcal{CN}(0, 2\sigma^2)$. **Magnitude/Phase Form:** $H = Re^{j\Theta}$

- Magnitude $R = \sqrt{X^2 + Y^2} \sim \text{Rayleigh}(\sigma^2)$
- Phase $\Theta = \tan^{-1}(Y/X) \sim \text{Uniform}[-\pi, \pi]$
- Joint: $f_{R, \Theta}(r, \theta) = \frac{r}{2\pi\sigma^2} e^{-r^2/2\sigma^2}$

6 Exam Problem Solving Strategies

6.1 Strategy: Finding Constants in PDF

Problem: Given $f_X(x) = c \cdot g(x)$ on support S . Find c . **Solution:**

Solve $\int_S c \cdot g(x) dx = 1$. *Example:* $f(x, y) = cxy$ on triangle.

$$\int_0^1 \int_0^x cxy dy dx = 1.$$

6.2 Strategy: Probability over Region

Problem: Find $P(X > Y)$ given f_{XY} . 1. **Draw** the support of f_{XY} .

2. **Draw** the event line (e.g., $y = x$). Shade region $y < x$. 3.

Integrate PDF over shaded region.

$$P(X > Y) = \int_{x_{\min}}^{x_{\max}} \int_{y_{\min}}^{\min(y_{\max}, x)} f_{XY}(x, y) dy dx$$

6.3 Strategy: Central Limit Theorem (CLT)

Let X_i be i.i.d with mean μ , var σ^2 . Let $Y = \sum_{i=1}^n X_i$. For large n ,

$$Y \sim N(n\mu, n\sigma^2). \quad \textbf{Standardization: } P(Y \leq k) \approx \Phi\left(\frac{k - n\mu}{\sqrt{n}\sigma}\right)$$

Continuity Correction (Discrete X_i): If X_i are integers (Bernoulli/Binomial/Poisson):

- $P(Y \leq k) \rightarrow P(Y_{\text{Gauss}} \leq k + 0.5)$
- $P(Y \geq k) \rightarrow P(Y_{\text{Gauss}} \geq k - 0.5)$
- $P(Y = k) \rightarrow P(k - 0.5 \leq Y_{\text{Gauss}} \leq k + 0.5)$

Exam Q3 Tip: $\mu = p, \sigma^2 = p(1 - p)$. $n = 400$.

6.4 Strategy: Transformation $Y = 1/|X|$

Given $f_X(x)$, find $f_Y(y)$ for $Y = 1/|X|$. 1. **Support:** If $-2 < x < 1$, then $y \in (1, \infty) \cup (0.5, \infty)$. 2. **CDF:**

$$F_Y(y) = P(1/|X| \leq y) = P(|X| \geq 1/y)$$

$$= P(X \geq 1/y) + P(X \leq -1/y)$$

$$= [1 - F_X(1/y)] + F_X(-1/y)$$

3. Differentiate with respect to y to get PDF.

7 Limit Theorems

$$\textbf{Markov: } X \geq 0 \Rightarrow P(X \geq a) \leq \frac{E[X]}{a}.$$

$$\textbf{Chebyshev: } P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

$$\textbf{WLLN: } \bar{X}_n \xrightarrow{p} \mu.$$

8 Math Reference

$$\textbf{Series: } \sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}; \quad \sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}$$

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x; \quad \sum_{k=1}^{\infty} kp(1 - p)^{k-1} = 1/p$$

$$\textbf{Integration: } \int x^n dx = \frac{x^{n+1}}{n+1}; \quad \int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1) \quad (\text{Integration by parts})$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (\text{Gaussian Integral})$$

9 Exam Solutions (AY23/24 & AY24/25)

AY23/24 Semester 1 Solutions

Q1: Dice ($A = \{4, 5, 6\}, p = 0.5$)

a) 3 A's in 10: Binom. $\binom{10}{3} (0.5)^{10} \approx 0.117$.

b) 5th is 3rd A: NegBin. (2 A in 4) then A. $\binom{4}{2} (0.5)^4 (0.5) = 0.188$.

c) Wait ≥ 4 : 1st 3 are A^c . $0.5^3 = 0.125$. **d) A before "1":**
 $\frac{P(A)}{P(A) + P(1)} = \frac{1/2}{1/2 + 1/6} = 0.75$.

Q2: Game (First to 3 wins, $p = 0.5$)

a) PMF (Total Rnds X): $P(3) = 2(0.5)^3 = 0.25$. $P(4) = 2\binom{3}{2} (0.5)^4 = 0.375$. $P(5) = 2\binom{4}{2} (0.5)^5 = 0.375$. $E[X] = 4.125$.

b) Cond: A wins given A won 1st $\rightarrow$ Need 2 wins in next 4. Binom sum $k \in \{2, 3, 4\} \rightarrow 0.6875$. **c) Bayes:** $P(W_1|W_A) = \frac{0.6875(0.5)}{0.5} = 0.6875$.

Q3: Triangle PDF (Symm @ 1): x ($0 \rightarrow 1$), $2 - x$ ($1 \rightarrow 2$).

a) CDF: $0.5x^2$ ($x < 1$); $1 - 0.5(2 - x)^2$ ($x \geq 1$). **b) $E[X]$** = 1 (Sym). $E[X^2] = 7/6 \rightarrow \text{Var} = 1/6$. **c) Trans** $Y = 1/X$: $F_Y(y) = 1 - F_X(1/y)$ for $y \geq 0.5$.

Q4: Joint $c(x + y)$ on $x(0, 2), y(0, 1)$:

a) $\iint c(x + y) = 1 \Rightarrow 3c = 1 \Rightarrow c = 1/3$. **b) $E[X^2]$** = 16/9. **c) $\text{Cov} = E[XY] - E[X]E[Y]$** = $2/3 - (10/9)(5/9) \approx -0.012$.

Q5: Shooting ($P_A = 0.4, P_B = 0.6$)

a) A win 2nd: $M_A M_B H_A \rightarrow 0.6(0.4)(0.4) = 0.096$. **b) A Win Game:** Geom Series $r = P(A^c)P(B^c) = 0.24$. $P = \frac{0.4}{1 - 0.24} \approx 0.526$.

c) Fair: $\frac{0.4}{1 - 0.6(1 - P_B)} = 0.5 \Rightarrow P_B = 2/3$.

Q6: Rounding ($X \sim U(-0.5, 7.5)$)

a) PMF: 8 bins width 1 $\rightarrow P(Y = k) = 1/8$. **b) Err Δ :** $\sim U(-0.5, 0.5)$. $E = 0, \text{Var} = 1/12$. **c) Sum ($n = 50$):** CLT. $\sigma_s^2 = 50/12$. $P(|Z| > \frac{3}{\sqrt{4.16}}) \approx 0.14$.

AY24/25 Semester 1 Solutions

Q1: Basics

1. Coin: $P(2H) = 0.36 \rightarrow p = 0.6$. Wait ≥ 5 ($TTTT \dots$) = 0.4^4 . Cond: (3H in 10) $\cap$ (1st H at 5). Fix $TTTTH$, need 2H in 5. $\binom{5}{2} / \binom{10}{3} = 1/12$.

2. Sets: $P(B|A) = 0.19$. $P(A) = P(AB) + P(AB^c)$. Solve $\frac{x}{x + 0.15} = 0.19 \rightarrow P(A) = 0.185$.

Q2: 10-Sided Dice ($P(1) \dots P(4)$ linear)

a) Indep: Check $P(X_1 \leq 3)P(X_2 > 2) = P(\text{Joint})$. $0.6 \times 0.7 = 0.42$.

b) $E[X]$ = 3, $\text{Var} = 1$. $\text{Var}(\text{Sum}) = 2$.

c) Sum ≥ 5 : $1 - P(\text{sum} < 5) = 1 - P\{(1, 1) \dots (3, 1)\} = 0.85$.

d) Cond: $\frac{P(X_1 \leq 2 \cap S \geq 5)}{0.85}$. Valid: (1, 4), (2, 3), (2, 4) $\rightarrow 0.21/0.85$.

Q3: CLT Binomial ($n = 400, p = 0.2$)

a) $\mu = 80, \sigma = 8$.

b) $P(63 < X \leq 88)$: CC $\rightarrow (63.5, 88.5)$. $Z \in (-2.06, 1.06)$. $1 - Q(2.06) - Q(1.06)$.

c) $P(X = 80)$: ± 0.5 range. $1 - 2Q(0.06)$.

Q4: Piecewise PDF (ax^2 on $-2..0$; bx on $0..1$)

a) Constants: $\int f = 1$ and $\int xf = -5/12$. Sys Eq $\rightarrow a = 3/16, b = 1$.

b) CDF: $0.5 + x^3/16$ (neg); $0.5 + x^2/2$ (pos).

c) $Y = 1/|X|$: $F_Y(y) = 1 - F_X(1/y) + F_X(-1/y)$.

Q5: Joint Triangle ($0 < y < 1 - x$)

a) Find c : $\int_0^1 \int_0^{1-x} cxy dy dx = 1 \Rightarrow c = 24$.

b) $P(Y > 0.5)$: $\int_{0.5}^1 f_Y(y) dy = 5/16$.

c) Cond $P(X > Y|Y < 0.5)$: Draw $y = x$. Split region. 8/11.

Found: <https://github.com/brianstm/NUS.git>